EA

ELITE IAL MATHEMATICS

EDEXCEL INTERNATIONAL A LEVEL

# WMA12/01 May/June 2024 Worked Solutions

May/June 2024 paper rebuilt with the worked solution after each question.

**10**
paper questions

**WMA12**
Pure 2

**Single-paper solutions**
standalone IAL route


**Dr Eslam Ahmed**



**Prepared for Dr Eslam Ahmed - eliteigcse.com**


P2

PAST PAPER

# WMA12/01 May/June 2024

**May/June 2024 | 10 questions | 75 marks**

| | | |
|---|---|---|
| **10** questions | **75** marks | **Answers** worked solution after each question |

5 marks

# Question 1

Binomial Expansion

**1.** (a) Find the first four terms, in ascending powers of $x$, of the binomial expansion of

$$\left(1-\frac{1}{6}x\right)^9$$

giving each term in simplest form.

**(3)**

(b) Hence find the coefficient of $x^3$ in the expansion of

$$(10x+3)\left(1-\frac{1}{6}x\right)^9$$

giving the answer in simplest form.

**(2)**

WMA12/01 MAY/JUNE 2024

# Worked Solution - Question 1

Binomial Expansion

Topic group

### 1. Expand the first terms

$$\left(1-\frac{1}{6}x\right)^9=1+9\left(-\frac{1}{6}x\right)+\binom{9}{2}\left(-\frac{1}{6}x\right)^2+\binom{9}{3}\left(-\frac{1}{6}x\right)^3+\cdots.$$

### 2. Simplify

The first four terms are $1-\frac{3}{2}x+x^2-\frac{7}{18}x^3$.

### 3. Identify contributions to x cubed

In $(10x+3)\left(1-\frac{1}{6}x\right)^9$, the $x^3$ coefficient comes from $3$ times the $x^3$ coefficient and $10x$ times the $x^2$ coefficient.

### 4. Calculate the coefficient

$$3\left(-\frac{7}{18}\right)+10(1)=-\frac{7}{6}+10=\frac{53}{6}.$$

**Final answer**

$(a)\ 1-\frac{3}{2}x+x^2-\frac{7}{18}x^3.$

$(b)\ \frac{53}{6}.$

# Question 2

7 marks

Arithmetic Sequences

**2.** **In this question you must show all stages of your working.**
**Solutions relying entirely on calculator technology are not acceptable.**

In an arithmetic series,

- the sixth term is 2
- the sum of the first ten terms is –80

For this series,

(a) find the value of the first term and the value of the common difference.

**(4)**

(b) Hence find the smallest value of $n$ for which

$$S_n > 8000$$

**(3)**

WMA12/01 MAY/JUNE 2024

# Worked Solution - Question 2

Arithmetic Sequences

Topic group

### 1. Use the sixth term

For an arithmetic series, $u_6 = a + 5d = 2$.

### 2. Use the sum of ten terms

$S_{10} = \frac{10}{2}(2a + 9d) = -80$, so $2a + 9d = -16$.

### 3. Solve simultaneously

From $a + 5d = 2$, $a = 2 - 5d$. Substitute into $2a + 9d = -16$.

### 4. Find a and d

$2(2 - 5d) + 9d = -16$, so $4 - d = -16$ and $d = 20$. Then $a = 2 - 5(20) = -98$.

### 5. Write S n

$S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(-196 + 20n - 20) = 10n^2 - 108n$.

### 6. Set up the inequality

We need $10n^2 - 108n > 8000$.

### 7. Solve the boundary

$10n^2 - 108n - 8000 = 0$ gives $5n^2 - 54n - 4000 = 0$.

### 8. Find the positive threshold

$$n = \frac{54 + \sqrt{82916}}{10} = 34.195\ldots.$$

### 9. Choose the smallest integer

Therefore the smallest integer value with $S_n > 8000$ is $n = 35$.

**Final answer**

$(a)$ First term $= -98$, common difference $= 20$.

$(b)$ $n = 35$.

# Question 3

6 marks

Laws of Logarithms

**3.** **In this question you must show all stages of your working.**
**Solutions relying entirely on calculator technology are not acceptable.**

(i) Using the laws of logarithms, solve

$$2\log_2(2-x) = 4 + \log_2(x+10)$$

**(5)**

(ii) Find the value of

$$\log_{\sqrt{a}} a^6$$

where $a$ is a positive constant greater than 1

**(1)**

WMA12/01 MAY/JUNE 2024

# Worked Solution - Question 3

Laws of Logarithms

Topic group

### 1. State the domain

The logarithms require $2 - x > 0$ and $x + 10 > 0$, so $-10 < x < 2$.

### 2. Combine powers

$2\log_2(2 - x) = \log_2((2 - x)^2)$.

### 3. Rewrite four

$4 = \log_2 16$, so $4 + \log_2(x + 10) = \log_2(16(x + 10))$.

### 4. Equate arguments

$(2 - x)^2 = 16(x + 10)$.

### 5. Solve the quadratic

$x^2 - 4x + 4 = 16x + 160$, so $x^2 - 20x - 156 = 0$.

### 6. Use the domain

$x = 26$ or $x = -6$. The value $26$ is outside the domain, so $x = -6$.

### 7. Evaluate the logarithm

$$\log_{\sqrt{a}}(a^6) = \log_{a^{1/2}}(a^6) = \frac{6}{1/2} = 12.$$

**Final answer**

$(i)$ $x = -6$.

$(ii)$ $12$.

WMA12/01 MAY/JUNE 2024

# Question 4

8 marks

Polynomials

**4.** $$f(x) = (x - 2)(2x^2 + 5x + k) + 21$$

where $k$ is a constant.

(a) State the remainder when $f(x)$ is divided by $(x - 2)$

**(1)**

Given that $(2x - 1)$ is a factor of $f(x)$

(b) show that $k = 11$

**(2)**

(c) Hence

(i) fully factorise $f(x)$,

(ii) find the number of real solutions of the equation

$$f(x) = 0$$

giving a reason for your answer.

**(5)**

WMA12/01 MAY/JUNE 2024

# Worked Solution - Question 4

Polynomials

Topic group

### 1. Use the remainder theorem

The remainder on division by $(x-2)$ is $f(2)$.

### 2. Find f of two

$f(2) = (2-2)(2(2)^2 + 5(2) + k) + 21 = 21$.

### 3. Use the factor theorem

Since $(2x-1)$ is a factor, $x = \frac{1}{2}$ is a root, so $f\left(\frac{1}{2}\right) = 0$.

### 4. Substitute x equals half

$$\left(\frac{1}{2} - 2\right)\left(2\left(\frac{1}{2}\right)^2 + 5\left(\frac{1}{2}\right) + k\right) + 21 = 0.$$

### 5. Solve for k

$-\frac{3}{2}(3+k) + 21 = 0$. Multiplying by $2$ gives $-3(k+3) + 42 = 0$, so $k = 11$.

### 6. Write f with k equals eleven

$f(x) = (x-2)(2x^2 + 5x + 11) + 21 = 2x^3 + x^2 + x - 1$.

### 7. Factorise

$2x^3 + x^2 + x - 1 = (2x-1)(x^2 + x + 1)$.

### 8. Count real solutions

The quadratic $x^2 + x + 1$ has discriminant $1 - 4 = -3$, so it has no real roots. Therefore $f(x) = 0$ has one real solution.

**Final answer**

$(a)$ $21$.

$(b)$ $k = 11$.

$(c)(i)$ $(2x - 1)(x^2 + x + 1)$.

$(c)(ii)$ One real solution.

# Question 5

6 marks

Proof

**5.** **In this question you must show detailed reasoning.**

(a) Given that $x$ and $y$ are positive numbers such that

$$(x-y)^3 > x^3 - y^3$$

prove that

$$y > x$$

**(4)**

(b) Using a counter example, show that the result in part (a) is not true for all real numbers.

**(2)**

WMA12/01 MAY/JUNE 2024

# Worked Solution - Question 5

Proof

Topic group

### 1. Start from the inequality

$(x-y)^3 > x^3 - y^3$.

### 2. Expand the left side

$x^3 - 3x^2y + 3xy^2 - y^3 > x^3 - y^3$.

### 3. Cancel common terms

Subtract $x^3 - y^3$ from both sides to get $-3x^2y + 3xy^2 > 0$.

### 4. Factor

$3xy(y-x) > 0$.

### 5. Use positivity

Since $x$ and $y$ are positive, $3xy > 0$. Therefore $y - x > 0$, so $y > x$.

### 6. Give a real-number counterexample

Take $x = 1$ and $y = -1$. Then $(x-y)^3 = 2^3 = 8$ and $x^3 - y^3 = 1 - (-1) = 2$, so the given inequality is true.

### 7. Show the conclusion fails

But $y > x$ would mean $-1 > 1$, which is false. Therefore the result is not true for all real numbers.

**Final answer**

$(a)$ Proven.

$(b)$ For example, $x = 1, y = -1$ satisfies the inequality but not $y > x$.

# Question 6

9 marks

Integration

**6.** (a) Sketch the curve with equation

$$y = a^x + 4$$

where $a$ is a positive constant greater than 1

On your sketch, show

- the coordinates of the point of intersection of the curve with the $y$-axis
- the equation of the asymptote of the curve

**(3)**

| $x$ | 2 | 2.3 | 2.6 | 2.9 | 3.2 | 3.5 |
|---|---|---|---|---|---|---|
| $y$ | 0 | 0.3246 | 0.8629 | 1.6643 | 2.7896 | 4.3137 |

The table shows corresponding values of $x$ and $y$ for

$$y = 2^x - 2x$$

with the values of $y$ given to 4 decimal places as appropriate.

Using the trapezium rule with all the values of $y$ in the given table,

(b) obtain an estimate for $\int_2^{3.5} \left(2^x - 2x\right) \mathrm{d}x$, giving your answer to 2 decimal places.

**(3)**

(c) Using your answer to part (b) and making your method clear, estimate

(i) $\int_2^{3.5} \left(2^x + 2x\right) \mathrm{d}x$

(ii) $\int_2^{3.5} \left(2^{x+1} - 4x\right) \mathrm{d}x$

**(3)**

# Worked Solution - Question 6

Integration

Topic group

### 1. Sketch key features

For $y = a^x + 4$ with $a > 1$, the curve is increasing.

### 2. Find the y-intercept

At $x = 0$, $y = a^0 + 4 = 5$, so the $y$-intercept is $(0, 5)$.

### 3. Find the asymptote

As $x \to -\infty$, $a^x \to 0$, so the horizontal asymptote is $y = 4$.

### 4. Use the trapezium width

The $x$ values increase by $0.3$, so $h = 0.3$.

### 5. Apply the trapezium rule

$$\int_2^{3.5} (2^x - 2x)dx \approx \frac{0.3}{2}[0 + 4.3137 + 2(0.3246 + 0.8629 + 1.6643 + 2.7896)]$$

.

### 6. Round part b

This gives $2.339475$, so the estimate is $2.34$ to 2 decimal places.

### 7. Estimate part c i

$2^x + 2x = (2^x - 2x) + 4x$, so $\int_2^{3.5} (2^x + 2x)dx \approx 2.34 + \int_2^{3.5} 4x\,dx$.

### 8. Add the extra integral

$\int_2^{3.5} 4x\,dx = [2x^2]_2^{3.5} = 16.5$, so the estimate is $2.34 + 16.5 = 18.84$.

**9. Estimate part c ii**

$2^{x+1} - 4x = 2(2^x - 2x)$, so the estimate is $2(2.34) = 4.68$.

**Final answer**

$(a)$ $y$-intercept $(0, 5)$ and asymptote $y = 4$.

$(b)$ $2.34$.

$(c)(i)$ $18.84$.

$(c)(ii)$ $4.68$.

WMA12/01 MAY/JUNE 2024

6 marks

# Question 7

Circles

**7.** The circle $C_1$ has equation

$$x^2 + y^2 + 8x - 10y = 29$$

(a) (i) Find the coordinates of the centre of $C_1$

(ii) Find the exact value of the radius of $C_1$

**(3)**

**In part (b) you must show detailed reasoning.**

The circle $C_2$ has equation

$$(x-5)^2 + (y+8)^2 = 52$$

(b) Prove that the circles $C_1$ and $C_2$ neither touch nor intersect.

**(3)**

WMA12/01 MAY/JUNE 2024

# Worked Solution - Question 7

Circles

Topic group

### 1. Complete the square

$x^2 + y^2 + 8x - 10y = 29$ becomes $(x+4)^2 - 16 + (y-5)^2 - 25 = 29$.

### 2. Write circle C1 in standard form

$(x+4)^2 + (y-5)^2 = 70$.

### 3. Read centre and radius

The centre of $C_1$ is $(-4, 5)$ and its radius is $\sqrt{70}$.

### 4. Read C2

$C_2$ has equation $(x-5)^2 + (y+8)^2 = 52$, so its centre is $(5, -8)$ and its radius is $\sqrt{52} = 2\sqrt{13}$.

### 5. Find distance between centres

The distance between $(-4, 5)$ and $(5, -8)$ is $\sqrt{9^2 + (-13)^2} = \sqrt{250} = 5\sqrt{10}$.

### 6. Compare with radii

$\sqrt{70} + 2\sqrt{13} = 15.58\ldots$, while $5\sqrt{10} = 15.81\ldots$.

### 7. Conclude

The distance between centres is greater than the sum of the radii, so the circles are separate. Therefore they neither touch nor intersect.

**Final answer**

$(a)(i)$ $(-4, 5)$.

$(a)(ii)$ $\sqrt{70}$.

$(b)$ The circles neither touch nor intersect.

WMA12/01 MAY/JUNE 2024

12 marks

# Question 8

Trigonometric Equations

**8.** **In this question you must show all stages of your working.**
**Solutions relying entirely on calculator technology are not acceptable.**

(i) Solve, for $0 < x \leqslant \pi$, the equation

$$5\sin x \tan x + 13 = \cos x$$

giving your answer in radians to 3 significant figures.

**(5)**

(ii) The temperature inside a greenhouse is monitored on one particular day.

The temperature, $H$°C, inside the greenhouse, $t$ hours after midnight, is modelled by the equation

$$H = 10 + 12\sin(kt + 18)° \qquad 0 \leqslant t < 24$$

where $k$ is a constant.

**Use the equation of the model to answer parts (a) to (c).**

Given that

- the temperature inside the greenhouse was 20 °C at 6 am
- $0 < k < 20$

(a) find all possible values for $k$, giving each answer to 2 decimal places.

**(4)**

Given further that $0 < k < 10$

(b) find the maximum temperature inside the greenhouse,

**(1)**

(c) find the time of day at which this maximum temperature occurs.

Give your answer to the nearest minute.

**(2)**

WMA12/01 MAY/JUNE 2024

# Worked Solution - Question 8

Trigonometric Equations

Topic group

### 1. Rewrite the trig equation

$5\sin x\tan x + 13 = \cos x$ becomes $\dfrac{5\sin^2 x}{\cos x} + 13 = \cos x$.

### 2. Multiply by cos x

$5\sin^2 x + 13\cos x = \cos^2 x$.

### 3. Use sine squared

$5(1-\cos^2 x) + 13\cos x = \cos^2 x$, so $6\cos^2 x - 13\cos x - 5 = 0$.

### 4. Solve for cos x

$(2\cos x - 5)(3\cos x + 1) = 0$, so $\cos x = \dfrac{5}{2}$ or $\cos x = -\dfrac{1}{3}$.

### 5. Choose the valid root

$\cos x = \dfrac{5}{2}$ is impossible, so $\cos x = -\dfrac{1}{3}$. For $0 < x \le \pi$, $x = 1.9106\ldots$, so $x = 1.91$ radians to 3 significant figures.

### 6. Use the temperature at 6 am

At $t = 6$, $20 = 10 + 12\sin(6k+18)^\circ$, so $\sin(6k+18)^\circ = \dfrac{5}{6}$.

### 7. Solve for k

$\sin^{-1}\left(\dfrac{5}{6}\right) = 56.442\ldots^\circ$. Since $0 < k < 20$, $18 < 6k + 18 < 138$, so $6k + 18 = 56.442\ldots$ or $123.557\ldots$.

### 8. Find possible k values

This gives $k = 6.407\ldots$ or $k = 17.592\ldots$, so $k = 6.41$ or $17.59$ to 2 decimal places.

**9. Use the tighter condition**

Given further that $0 < k < 10$, use $k = 6.407\ldots$.

**10. Find the maximum temperature**

The maximum value of $\sin$ is $1$, so the maximum temperature is $10 + 12 = 22°\mathrm{C}$.

**11. Find the time of maximum**

The maximum occurs when $kt + 18 = 90$, so $t = \dfrac{72}{6.407\ldots} = 11.2375\ldots$ hours after midnight.

**12. Convert to time of day**

$0.2375\ldots$ hours is $14.25\ldots$ minutes, so the maximum occurs at about $11{:}14$ am.

**Final answer**

$(i)$ $x = 1.91$ radians.

$(ii)(a)$ $k = 6.41$ or $17.59$.

$(ii)(b)$ $22°\mathrm{C}$.

$(ii)(c)$ $11{:}14$ am.

# Question 9



**9.**

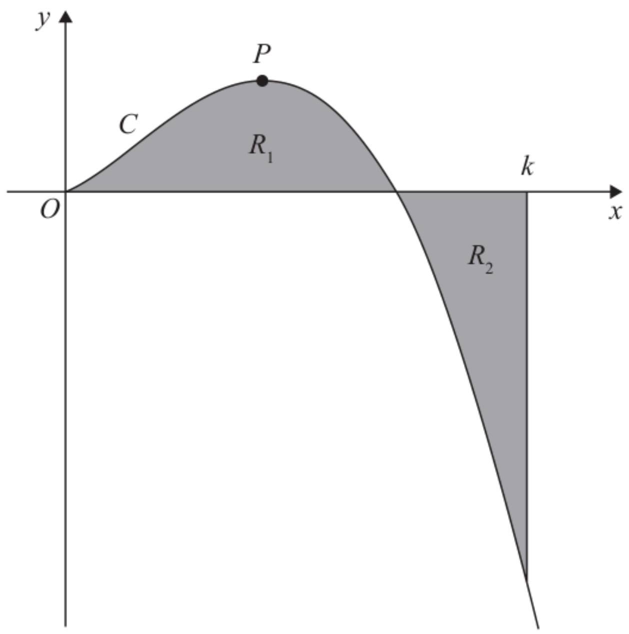


**Figure 1**

Figure 1 is a sketch of the curve $C$ with equation

$$y = 2x^{\frac{3}{2}}(4 - x) \qquad x \geqslant 0$$

The point $P$ is the stationary point of $C$.

(a) Find, using calculus, the $x$ coordinate of $P$.

**(4)**

The region $R_1$, shown shaded in Figure 1, is bounded by $C$ and the $x$-axis.

The region $R_2$, also shown shaded in Figure 1, is bounded by $C$, the $x$-axis and the line with equation $x = k$, where $k$ is a constant.

Given that the area of $R_1$ is equal to the area of $R_2$

(b) find, using calculus, the exact value of $k$.

**(4)**

WMA12/01 MAY/JUNE 2024

# Worked Solution - Question 9

Applications of Differentiation

Topic group

**1. Rewrite the curve**

$y = 2x^{3/2}(4 - x) = 8x^{3/2} - 2x^{5/2}$.

**2. Differentiate**

$\dfrac{dy}{dx} = 12x^{1/2} - 5x^{3/2} = \sqrt{x}(12 - 5x)$.

**3. Find the stationary point**

For the interior stationary point $P$, $12 - 5x = 0$, so $x = \dfrac{12}{5}$.

**4. Set up the area condition**

$R_1$ is the area from $x = 0$ to $x = 4$ above the $x$-axis. $R_2$ is the area from $x = 4$ to $x = k$ below the $x$-axis.

**5. Use equal areas**

Equal areas mean $\displaystyle\int_0^4 y\,dx = -\int_4^k y\,dx$, so $\displaystyle\int_0^k y\,dx = 0$.

**6. Integrate y**

$\displaystyle\int 2x^{3/2}(4 - x)dx = \int (8x^{3/2} - 2x^{5/2})dx = \frac{16}{5}x^{5/2} - \frac{4}{7}x^{7/2}$.

**7. Solve for k**

$\dfrac{16}{5}k^{5/2} - \dfrac{4}{7}k^{7/2} = 0$.

**8. Factor**

$4k^{5/2}\left(\dfrac{4}{5} - \dfrac{k}{7}\right) = 0$.

### 9. Choose the value beyond four

Since $k > 4$, $\frac{4}{5} - \frac{k}{7} = 0$, so $k = \frac{28}{5}$.

**Final answer**

$(a)\ x = \frac{12}{5}$.

$(b)\ k = \frac{28}{5}$.

# Question 10

8 marks

Modelling with Sequences & Series

**10.** **In this question you must show all stages of your working.**
**Solutions relying entirely on calculator technology are not acceptable.**

The number of dormice and the number of voles on an island are being monitored.

Initially there are 2000 dormice on the island.

A model predicts that the number of dormice will increase by 3% each year, so that the numbers of dormice on the island at the end of each year form a geometric sequence.

(a) Find, according to the model, the number of dormice on the island 6 years after monitoring began. Give your answer to 3 significant figures.

**(2)**

The number of voles on the island is being monitored over the same period of time.

Given that

- 4 years after monitoring began there were 3690 voles on the island
- 7 years after monitoring began there were 3470 voles on the island
- the number of voles on the island at the end of each year is modelled as a geometric sequence

(b) find the equation of this model in the form

$$N = ab^t$$

where $N$ is the number of voles, $t$ years after monitoring began and $a$ and $b$ are constants. Give the value of $a$ and the value of $b$ to 2 significant figures.

**(3)**

When $t = T$, the number of dormice on the island is equal to the number of voles on the island.

(c) Find, according to the models, the value of $T$, giving your answer to one decimal place.

**(3)**

WMA12/01 MAY/JUNE 2024

# Worked Solution - Question 10

Modelling with Sequences & Series

Topic group

### 1. Model the dormice

The dormice model is $D = 2000(1.03)^t$.

### 2. Find the dormice after six years

$D = 2000(1.03)^6 = 2388.10\ldots$, so there are $2390$ dormice to 3 significant figures.

### 3. Set up the vole model

Let the vole model be $N = ab^t$.

### 4. Use the two data points

At $t = 4$, $ab^4 = 3690$. At $t = 7$, $ab^7 = 3470$.

### 5. Find b

Dividing gives $b^3 = \dfrac{3470}{3690}$, so $b = 0.9797\ldots$, which is $0.98$ to 2 significant figures.

### 6. Find a

$a = \dfrac{3690}{b^4} = 4005.18\ldots$, which is $4000$ to 2 significant figures.

### 7. Write the vole model

So the model is approximately $N = 4000(0.98)^t$.

### 8. Set the two models equal

Use $2000(1.03)^T = ab^T$ with the unrounded constants from the vole model.

### 9. Solve using logarithms

$$\left(\frac{1.03}{b}\right)^T = \frac{a}{2000}, \text{ so } T = \frac{\log(a/2000)}{\log(1.03/b)}.$$

### 10. Calculate T

Substituting $a = 4005.18\ldots$ and $b = 0.9797\ldots$ gives $T = 13.875\ldots$, so $T = 13.9$ to one decimal place.

**Final answer**

$(a)$ $2390$ dormice.

$(b)$ $N = 4000(0.98)^t$ to 2 s.f. constants.

$(c)$ $T = 13.9$.